"Spatio-Temporal Traffic Prediction System Using CTNS"

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ABSTRACT

- Introduces an AI-Powered Driving Assistance System for Intelligent Traffic Management (ITMS).
- Uses computer vision, deep learning, and real-time analytics to optimize traffic flow and improve road safety.
- Features include YOLO-based vehicle detection, OpenCV-powered pothole detection, and emergency vehicle prioritization using color segmentation.
- AI-powered Driving Assistant offers speech recognition for real-time hazard alerts, benefiting all drivers, including those with hearing impairments.

PROBLEM STATEMENT

- Urban Traffic Challenges: Congestion, road safety risks, and emergency delays.
- Negative Impacts: Increased travel time, fuel consumption, and pollution.
- Pothole Hazards: Cause accidents and road damage.

PROPOSED SOLUTION

- ✓ Vehicle & Traffic Density Detection for real-time flow optimization.
- ✓ Live Pothole Detection (OpenCV) to prevent road damage & accidents.
- ✓ Emergency Vehicle Prioritization for faster response times.
- ✓ AI-Powered Driving Assistant (Speech Recognition) for hazard alerts.

METHODS & MATERIALS

METHODS

- Traffic Density Detection using YOLO for real-time monitoring.
- Pothole Detection with OpenCV for road damage prevention.
- Emergency Vehicle Prioritization via color segmentation for quick response.
- AI-Powered Driving Assistant using TensorFlow.js & Teachable Machine for hazard alerts.

MATERIALS

- HD/4K Cameras
- OpenCV for Computer Vision
- YOLO for Object Detection
- TensorFlow.js for Speech Recognition
- HD/4K Cameras

MODULE1:POTHOLE DETECTION

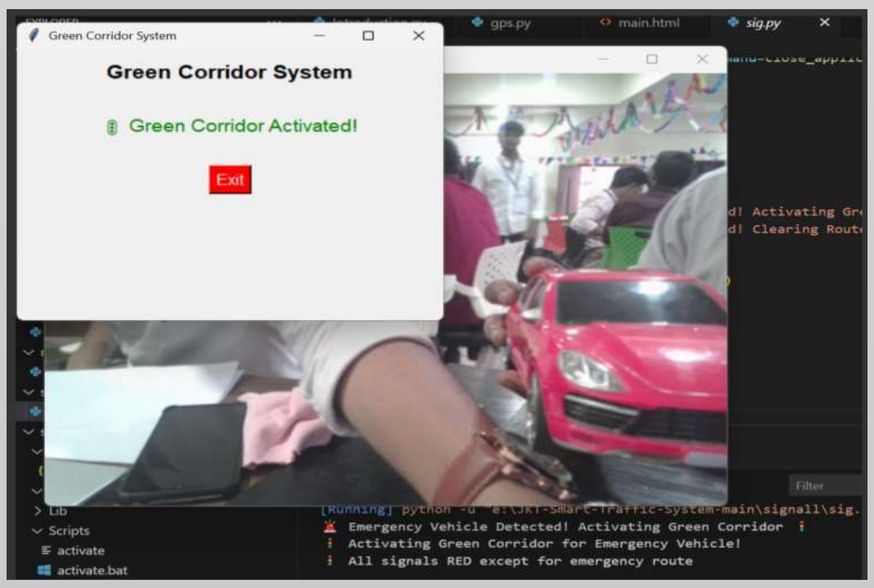
Safety Alerts: Displays bounding boxes and labels potholes to alert drivers and road maintenance teams.

MODULE2:TRAFFIC DETECTION

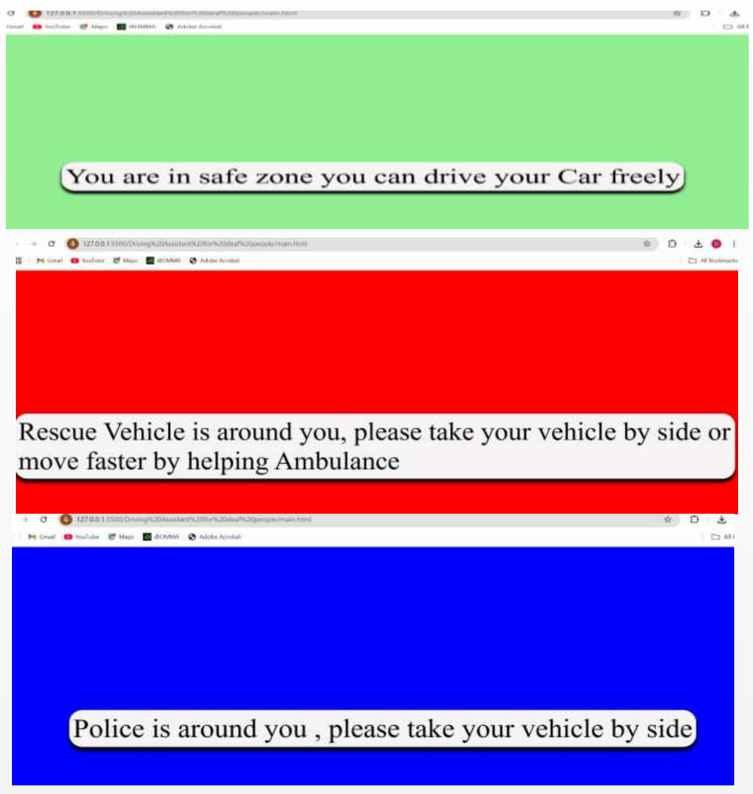
MODULE 3 : VEHICLE DETECTION

Traffic Optimization: Identifies vehicle types to improve road safety and traffic management.

MODULE 4:GREEN CORRIDOR SYSTEM



M5 : SPEECH RECOGNITION



REVENUE MODEL

- Subscription Model – SaaS-based AI traffic management system for government agencies & smart cities.
- Direct Sales – Licensing AI-powered traffic control software & hardware to municipalities.
- Ad-Based Model – Location-based traffic insights for businesses, enabling targeted advertising.
- Partnerships – Collaborate with navigation apps (Google Maps, Uber) & automotive companies.

COMPARISION WITH MODELS

1. Existing Smart Traffic Systems (like those in Singapore, the US, and Germany) use IoT, AI, and smart sensors integrated into city traffic networks.
2. This is focused more on computer vision-based detection, using AI models to analyze traffic density, classify vehicles, and detect road defects.
3. Lack of physical integration – Unlike existing smart traffic light systems (e.g., in the Netherlands), this project does not control traffic signals directly.

CONCLUSION

- Enhances urban mobility through AI-driven traffic management.
- Improves road safety by identifying hazards and prioritizing emergency vehicles.

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